

Do Not Anonymize Away the Answer: Measuring Task-Critical Meaning Loss in Text De-identification

Anonymous authors

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Abstract

Text de-identification is often evaluated as a span removal problem: did a system remove names, dates, locations, contact information, and other explicit identifiers? For sensitive text used in foundation-model workflows, this is necessary but incomplete. A transformed clinical vignette or legal case excerpt can pass a privacy-oriented screen while removing the medication dose, lab value, statutory article, procedural posture, or legal claim needed for downstream reasoning. We study this failure mode through a small clinical/legal benchmark for utility-preserving anonymization. The benchmark pairs privacy labels with task-critical facts and downstream QA items. We introduce *Task-Critical Fact Retention* (TCFR), a fact-level utility metric, and Critical Span Guard, a low-cost protocol that extracts privacy spans and important domain facts, anonymizes under preservation constraints, verifies the result, and applies a deterministic safety/repair layer. In a 150-example non-oracle LLM run, Critical Span Guard achieves zero measured direct identifier leaks, lower quasi-identifier risk (0.13 versus 1.73 for generic prompting and 0.57 for privacy-first prompting), and higher audited task-critical fact retention (0.99 versus 0.94 and 0.53). A handwritten stress slice further shows why the problem is not reducible to “redact more”: some sensitive facts, such as pregnancy, sexual orientation, domestic violence, or minority status, can also be the task-critical answer. The results support evaluating de-identification systems by both privacy protection and retained domain meaning.

1 Introduction

Sensitive clinical and legal text is valuable for retrieval-augmented generation, supervised fine-tuning, model evaluation, safety auditing, and synthetic-data generation. The usual first step before such use is de-identification, anonymization, or pseudonymization. However, a system optimized only for removing personally identifying information can destroy the very facts that make the text useful. In a clinical note, replacing “methotrexate 15 mg weekly” with “a medication” may preserve broad topical similarity while breaking the answer to a treatment question. In a legal excerpt, replacing “Article 6” with “a specified article” removes the legal basis needed to reason about the claim. These are not cosmetic changes: they alter the information that downstream models or evaluators are expected to use.

This tension is especially important for foundation-model data pipelines. A release pipeline that over-redacts data may be privacy-conservative but unable to support clinical decision-support evaluation, legal QA, or domain-specific retrieval. A pipeline that preserves all clinically or legally meaningful facts may leave quasi-identifiers, rare events, or sensitive attributes that can aid re-identification. In practice, data stewards must make policy choices about which facts are safe to preserve, which facts must be generalized, and which facts are too sensitive to release even when they are task-relevant. Yet many evaluations still reduce de-identification to span-level recall over predefined identifier categories.

We ask a practical question: *after anonymization, is the transformed text still useful for the domain task it was meant to support?* We focus on two domains where the privacy–utility tension is concrete. Clinical-style vignettes contain diagnoses, medications, doses, labs, symptoms,

procedures, and timelines. Public legal snippets derived from the Text Anonymization Benchmark (TAB) contain application identifiers, party roles, claims, legal articles, procedural facts, confidential attributes, and outcomes [Pilan et al. \(2022\)](#). Existing tools such as Microsoft Presidio [Microsoft \(2026\)](#) and regulatory frameworks such as HIPAA de-identification guidance [HHS \(2012\)](#) are essential privacy infrastructure, but they do not by themselves answer whether the remaining text preserves task-critical meaning.

This paper makes four contributions:

- We construct a 150-example controlled benchmark with synthetic clinical vignettes and TAB-derived legal snippets, each annotated with direct identifiers, quasi-identifiers, task-critical facts, and downstream QA items.
- We propose *Task-Critical Fact Retention* (TCFR), a fact-level utility metric that distinguishes retained domain meaning from generic surface similarity.
- We introduce Critical Span Guard, a simple anonymization protocol that extracts privacy spans and task-critical facts, anonymizes under explicit preservation constraints, and applies deterministic safety/repair rules for direct identifiers and known domain-specific failure modes.
- We provide a bounded empirical audit showing that generic prompting tends to preserve utility but leak identifiers, privacy-first prompting removes identifiers but often removes the answer, and task-aware structure improves the observed privacy–utility frontier.

We argue that span-level privacy evaluation alone can both miss residual risk and reward over-redaction. For sensitive text used in foundation-model workflows, a more useful evaluation must ask two questions jointly: what identifying information remains, and what domain meaning has been lost?

2 Related Work

Text de-identification. Clinical de-identification has a long history in shared tasks and patient-note benchmarks. Early neural systems framed the problem as sequence labeling over protected-health-information spans [Dernoncourt et al. \(2016\)](#), while annotation resources from i2b2 and related challenges supported systematic PHI recognition and normalization [Stubbs et al. \(2015\)](#). Recent work has explored stronger neural and LLM-based de-identification, but the dominant evaluation mode is still identifier detection or span replacement. This is a necessary component of privacy evaluation, but it does not directly measure whether clinically material facts such as drugs, doses, lab values, and procedures remain intact.

Legal anonymization. TAB provides an important benchmark for legal-text anonymization by annotating European Court of Human Rights cases with entity spans, masking decisions, confidential attributes, and coreference relations [Pilan et al. \(2022\)](#). Later work on neural legal anonymization further shows that legal documents require more than generic named-entity recognition because case identifiers, party roles, procedural details, and confidential attributes interact with legal meaning [Papadopoulou et al. \(2023\)](#). Our legal split builds on TAB but adds a utility target: legal article references, claims, outcomes, and procedural facts should be preserved when they are necessary for downstream reasoning.

Privacy risk beyond direct identifiers. Classical privacy work distinguishes direct identifiers from quasi-identifiers: attributes that may not identify a person alone but can do so in combination [Sweeney \(2002\)](#); [Narayanan & Shmatikov \(2008\)](#). Differential privacy gives a formal framework for aggregate releases and learning algorithms [Dwork et al. \(2006\)](#), but most practical text anonymization pipelines do not provide such guarantees. Recent LLM privacy work shows that powerful models can infer personal attributes from seemingly innocuous text [Staab et al. \(2023\)](#). Recent anonymization benchmarks therefore increasingly consider residual re-identification risk and utility together [Loiseau et al. \(2025\)](#); [Krčo et al. \(2026\)](#), and contemporaneous adaptive anonymization work treats the privacy–utility tradeoff as a task-specific prompt-optimization problem [Loiseau et al. \(2026\)](#). We share this motivation, but make the utility target explicit at the level of domain facts and QA answers.

96 **Utility and faithfulness metrics.** Generic text-generation metrics such as token overlap,
 97 embedding similarity, or BERTScore Zhang et al. (2020) can be useful diagnostics, but they
 98 do not encode what domain facts must be preserved. High similarity to the original can
 99 even be a warning sign in anonymization because identifiers and quasi-identifiers may
 100 remain. Conversely, low similarity may be acceptable if the transformed text preserves the
 101 clinically or legally relevant answer. This motivates a fact-level metric rather than relying
 102 only on surface similarity.

103 3 Benchmark and Task

104 **Task definition.** Given an input text x , a de-identification method outputs transformed text
 105 $\hat{x}$. Each example contains three types of annotation: direct identifiers D , quasi-identifiers Q ,
 106 and task-critical facts F . Direct identifiers include names, contact information, exact dates,
 107 medical record numbers, legal application numbers, and named institutions tied to a per-
 108 son. Quasi-identifiers include age, occupation, city, nationality, sensitive attributes, family
 109 relations, rare events, and combinations of details that may enable linkage. Task-critical
 110 facts include information needed to answer a domain question: clinical diagnoses, labs,
 111 medications, doses, symptoms, procedures, timelines, or legal articles, claims, procedures,
 112 outcomes, and reasoning. A good output should remove or generalize D and high-risk Q
 113 while preserving the facts in F at the specificity required by the downstream task.

114 **Clinical split.** We generate 75 synthetic clinical vignettes from templates. Each example
 115 contains direct identifiers such as patient name, clinician name, clinic, date, medical record
 116 number, phone number, and email; quasi-identifiers such as age, city, and occupation; and
 117 task-critical facts such as diagnosis, medication, dose, lab result, symptom duration, imaging
 118 result, or treatment plan. The clinical split is synthetic by design: it provides controlled
 119 utility labels without exposing real protected health information. It should be interpreted as
 120 a controlled stress test, not as evidence of performance on real clinical notes.

121 **Legal split.** We sample 75 short excerpts from TAB, an open-source corpus of English-
 122 language European Court of Human Rights cases annotated with personal identifiers,
 123 masking decisions, confidential attributes, and coreference relations Pilan et al. (2022). For
 124 each sampled snippet, we keep short procedure, complaint, and outcome sentences when
 125 available. We use TAB spans for privacy labels and heuristic extraction plus manual spot
 126 checks for legal articles, claims, outcomes, and QA answers. This lets us study legal meaning
 127 without processing private legal files.

128 **Stress slice.** We also construct a 12-example handwritten diagnostic slice that stresses
 129 cases where privacy and utility overlap: pregnancy, sexual orientation, domestic violence,
 130 Roma marriage, exact ages, rare occupations, case labels, secret evidence, exact legal arti-
 131 cle/protocol specificity, numeric labs, and dose preservation.

132 4 Metrics

133 We report privacy and utility metrics together. *Direct leak* is the fraction of rows where any
 134 gold direct identifier remains in the anonymized text. *QI risk* is the mean quasi-identifier
 135 risk score, where retained specific dates, locations, occupations, sensitive attributes, or
 136 combinations of details increase the risk. We also report direct, quasi, and all-span privacy
 137 recall: the fraction of gold privacy spans not found verbatim after anonymization.

138 For utility, exact TCFR measures the fraction of gold task-critical facts retained by determin-
 139 istic phrase or alias matching:

$$\text{TCFR}(\hat{x}) = \frac{1}{|F|} \sum_{f \in F} \mathbb{1}[f \text{ is retained in } \hat{x}]. \quad (1)$$

140 Exact matching is reproducible but conservative because faithful paraphrases may not
 141 match the gold string. We therefore also report audited TCFR using a structured LLM fact

judge that marks each fact as preserved, generalized-but-acceptable, omitted, contradicted, or unclear. The judge is strict for medication names, doses, lab values, legal Article numbers, outcomes, and legal claims. *QA consistency* asks whether downstream questions can still recover the gold answers from the anonymized text. Finally, we report surface metrics such as token F1, TF-IDF cosine, and edit rate only as diagnostic foils, not as privacy or task-utility guarantees.

5 Methods

We compare three non-oracle LLM anonymizers in the 150-example run. All methods operate on the same inputs and are evaluated with the same gold privacy and utility annotations.

Generic LLM. The generic prompt asks the model to remove personally identifying information while preserving document meaning. It resembles the simple instruction a data steward might give to a general-purpose model. This baseline tests whether broad de-identification instructions are sufficient.

Privacy-first LLM. The privacy-first prompt asks the model to remove or generalize any information that could identify a person, including rare locations, exact dates, named institutions, rare occupations, unique events, and combinations of details. It is intentionally conservative. This baseline tests the cost of resolving uncertainty by redacting or generalizing aggressively.

Critical Span Guard. Critical Span Guard adds structure. First, an extraction prompt identifies privacy spans and task-critical domain facts. Second, an anonymization prompt receives these structures and is instructed to replace direct identifiers with typed placeholders, generalize quasi-identifiers when possible, preserve task-critical facts, and use the least identifying version of facts that are both privacy-relevant and task-critical. For clinical text, the prompt explicitly protects medications, doses, labs, diagnoses, symptoms, imaging findings, and meaningful timelines. For legal text, it protects legal article/statute numbers, claims, procedural posture, outcomes, and reasoning, while treating legal case/application numbers as direct identifiers. Third, a verification/repair prompt checks for leaked direct identifiers, generalizable quasi-identifiers, omitted facts, contradictions, and invented facts. Finally, a deterministic safety/repair layer replaces direct identifiers such as case/application numbers, dates, MRNs, phone numbers, emails, names, and institutions, and repairs known domain mistakes such as legal Article over-generalization and medication placeholdering.

Diagnostic local baselines. We also run local diagnostic baselines on the original benchmark, a 200-example local expansion, and the stress slice: regex rules, Presidio, direct-span oracle replacement, privacy-first oracle redaction, and a CSG oracle using gold benchmark annotations. These oracle variants are not deployable systems; they diagnose upper bounds and failure modes.

6 Experimental Setup

The promoted non-oracle run contains 150 examples: 75 synthetic clinical vignettes and 75 TAB-derived legal snippets. The LLM methods are run with a GPT model. The exact model is not central to our claim: the paper evaluates prompt protocols and safety layers under a bounded budget rather than training a new model. We use no real protected health information. Clinical examples are synthetic, and legal examples are public TAB-derived snippets.

We compute deterministic privacy and exact-utility metrics from gold spans and facts. We then run an LLM-assisted fact-retention audit over the anonymized outputs. Paired deltas are computed over the same examples to compare Critical Span Guard against each baseline. We also inspect qualitative failures and maintain explicit claim boundaries: the privacy

Method	Direct leak ↓	QI risk ↓	Exact TCFR ↑	Audited TCFR ↑	QA ↑
Generic LLM	0.18	1.73	0.81	0.94	0.89
Privacy-first LLM	0.00	0.57	0.40	0.53	0.41
Critical Span Guard	0.00	0.13	0.88	0.99	0.97

Table 1: Main 150-example non-oracle results. Generic prompting preserves many facts but leaks identifiers; privacy-first prompting protects direct identifiers but often removes the answer; Critical Span Guard improves the measured privacy-utility frontier.

Domain	Method	Direct leak	QI risk	Exact TCFR	Audited TCFR	QA
Clinical	Generic	0.03	2.03	0.70	0.93	0.85
Clinical	Privacy-first	0.00	1.07	0.19	0.46	0.23
Clinical	Critical Span Guard	0.00	0.00	0.82	1.00	1.00
Legal	Generic	0.33	1.44	0.92	0.94	0.92
Legal	Privacy-first	0.00	0.08	0.60	0.60	0.60
Legal	Critical Span Guard	0.00	0.25	0.95	0.99	0.95

Table 2: Domain split for the 150-example non-oracle run. The legal split exposes the hardest residual tradeoff: privacy-first redaction attains lower QI risk by deleting or generalizing legally meaningful details, while Critical Span Guard preserves more legal utility but retains a small number of task-overlapping quasi-identifiers.

190 metrics are string/span and heuristic risk audits, not k-anonymity, differential privacy, legal
191 compliance, or protection against an adversary with external databases.

192 7 Results

193 Table 1 shows the core privacy-utility tradeoff. Generic prompting preserves many task
194 facts, but leaks direct identifiers in 27 of 150 examples and has high QI risk. Privacy-first
195 prompting removes all measured direct identifiers, but exact TCFR falls to 0.40 and QA
196 consistency to 0.41. Critical Span Guard achieves zero measured direct leaks, the lowest
197 overall QI risk, audited TCFR of 0.99, and QA consistency of 0.97.

198 The domain split in Table 2 reveals different failure modes. In clinical examples, generic
199 prompting leaves nearly all quasi-identifiers, whereas privacy-first prompting often replaces
200 specific diagnoses, drugs, labs, or doses with vague placeholders. Critical Span Guard
201 removes all measured clinical QI hits while retaining almost all audited facts. In legal
202 examples, the frontier is harder. Privacy-first prompting has lower QI risk than Critical
203 Span Guard on the legal subset because it generalizes legal claims, jurisdictions, statutes, or
204 sensitive attributes. This improves the measured privacy score but damages the legal utility
205 target.

206 The span-level privacy audit agrees with the row-level picture. Critical Span Guard reaches
207 direct span recall of 1.00 and all privacy-span recall of 0.98, compared with 0.92 and 0.77 for
208 generic prompting and 1.00 and 0.96 for privacy-first prompting. Paired deltas on the same
209 examples sharpen this interpretation. Relative to generic prompting, Critical Span Guard
210 reduces direct-leak rate by 0.18 and QI risk by 1.61; on legal rows alone, the QI reduction
211 is 1.19. Relative to privacy-first prompting, it improves exact TCFR by 0.49, audited TCFR
212 by 0.47, and QA consistency by 0.56. The domain-level audited-TCFR gains over privacy-
213 first are 0.54 on clinical rows and 0.39 on legal rows. The legal-only QI comparison is an
214 important caveat: Critical Span Guard is slightly worse than privacy-first prompting on legal
215 QI risk because it intentionally preserves some legal facts that overlap with quasi-identifying
216 content.

217 As a sample-size diagnostic, we also ran a bounded 300-example diagnostic expansion,
218 adding 75 clinical and 75 legal rows beyond the promoted n150 run. The qualitative frontier
219 persists: Critical Span Guard has zero measured direct leaks, QI risk 0.21, and audited TCFR

Method	n150 audit	GPT-5.5 fact60	Full-n150	Hard20 severe	Fixed30 audit	GPT-5.5 anon n30
Generic LLM	0.94	0.87	0.90	1.00	0.84	0.81
Privacy-first LLM	0.53	0.48	0.41	0.70	0.51	0.42
Critical Span Guard	0.99	0.92	0.92	0.90	0.97	0.89

Table 3: Follow-up robustness checks. Columns report audited TCFR except Hard20 severe, which is the GPT-5.5 adversarial privacy severe-flag rate on the 20 hardest rows. Full-n150 is completed model-auditor coverage for the promoted n150 outputs, but is not expert/human annotation. Fixed30 is a deliberately hard manual-style subset; GPT-5.5 anon n30 is a bounded stronger-model screen, not a broad model-scaling result.

Stage	Direct leak	QI risk	Exact TCFR	QA
CSG draft	0.07	1.03	0.88	0.97
CSG verified	0.07	1.03	0.88	0.97
CSG + safety/repair	0.00	0.13	0.88	0.97

Table 4: Critical Span Guard ablation. The verification prompt did not change deterministic metrics in this sample; the measured gain comes from the deterministic safety/repair layer.

0.99, compared with 0.19, 1.79, and 0.93 for generic prompting and 0.00, 0.62, and 0.52 for privacy-first prompting. On the added n200-to-n300 100-example slice alone, Critical Span Guard has zero direct leak rows, 14 QI-hit rows, and audited TCFR 0.99. We keep n150 as the promoted headline because the manuscript tables and completed manual-audit plan remain anchored there; the n300 downstream bundle is synchronized but its second-annotator packet has not yet been completed. Thus n300 is sample-size robustness evidence, not a promoted headline replacement.

To test judge dependence, we also ran an external GPT-5.5 fact-retention audit over all promoted n150 outputs. The stronger judge is stricter than the low-cost judge but preserves the utility ordering: audited TCFR is 0.92 for Critical Span Guard, 0.90 for generic prompting, and 0.41 for privacy-first prompting. The paired Critical Span Guard-minus-generic audited-TCFR gain is small and statistically uncertain (0.02), while the same rows still show generic direct leakage (0.18) and QI risk (1.73) versus zero direct leakage and QI risk 0.13 for Critical Span Guard. Fact-label agreement with the original audit is 0.92, 0.95, and 0.83 respectively, with most disagreements caused by strict clinical numeric or legal specificity. A GPT-5.5 adversarial privacy audit on the 20 hardest rows is similarly conservative: generic prompting has direct privacy flags on 60% of rows, Critical Span Guard on 5%, and privacy-first prompting on 0%, but privacy-first is the same method that loses most task-critical facts. We treat these external audits as robustness checks, not as expert annotation. Table 3 combines the follow-up checks used for claim triangulation. The bounded n30 GPT-5.5 anonymization screen shows that stronger generic prompting can remove obvious direct leaks on selected stress cases, but privacy-first prompting still loses utility and Critical Span Guard remains the best audited-TCFR corner. Critical Span Guard does not have the lowest upgraded QI risk on that stress screen, so we treat the result as reviewer-risk triage rather than a model-scaling claim.

Figure 1 summarizes the main privacy-utility frontier: the task-aware protocol moves toward lower QI risk without collapsing audited fact retention.

Table 4 is important for claim discipline. In this sample, the verification/repair prompt did not change deterministic metrics: the draft and verified rows are numerically identical. The measured gain comes from the deterministic safety/repair layer, which closes eleven direct leaks, reduces QI-risk rows from 74 to 11, repairs legal Article specificity losses, and leaves eleven legal residual cases. This result suggests that LLM-generated anonymization should be paired with explicit deterministic checks, not trusted as a single-pass transformation.

Table 5 addresses whether generic utility metrics are enough. Generic prompting and Critical Span Guard have nearly identical token F1 against the original text, and generic prompting has higher TF-IDF cosine similarity, but it also has far higher privacy failure and

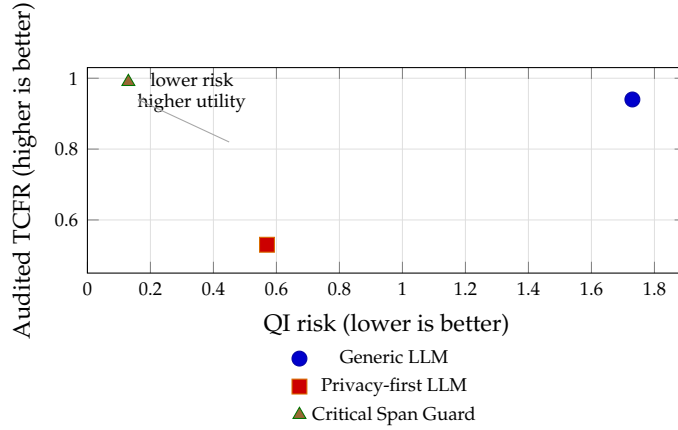


Figure 1: Privacy-utility frontier for the 150-example non-oracle LLM run. Critical Span Guard is closest to the upper-left target region: lower measured quasi-identifier risk and higher audited task-critical fact retention.

Method	Token F1	TF-IDF	Edit rate	Audited fact-fail rate	Privacy-fail rate
Generic LLM	0.78	0.71	0.25	0.24	0.75
Privacy-first LLM	0.58	0.47	0.46	0.79	0.27
Critical Span Guard	0.77	0.62	0.27	0.02	0.03

Table 5: Surface similarity metrics are not sufficient proxies for privacy-safe utility.

audited fact failure rates. Among the 225 method-example rows at or above the median TF-IDF cosine score, 73 still have privacy failures and 34 still have audited fact loss. Surface similarity therefore rewards text that may be too close to the original and misses task-critical failures; this is consistent with broader concerns that generic text-generation metrics do not encode domain-specific faithfulness by themselves [Zhang et al. \(2020\)](#).

The n200 diagnostic expansion in Table 6 gives the same qualitative frontier. Regex and direct-span replacement preserve exact utility but leave high QI risk. Presidio is useful infrastructure but performs poorly on our legal snippets because the benchmark contains legal identifiers and confidential attributes that are not standard clinical or generic PII. The privacy-first oracle nearly eliminates measured QI risk but loses more than half of task-critical facts. The CSG oracle indicates that a gold-informed version of task-aware anonymization can preserve utility while substantially reducing QI risk, though it remains an upper-bound diagnostic rather than a deployable system. The domain split explains the overall Presidio row: Presidio’s legal direct-leak rate is 0.98, while its clinical TCFR is only 0.45 under exact matching.

The stress slice in Table 7 clarifies the frontier. Privacy-first oracle redaction drives measured direct leaks and QI risk to zero, but fails QA on 11 of 12 rows. Direct-span replacement preserves utility but leaves quasi-identifiers in every row. CSG oracle replacement preserves all task facts with zero direct leaks, but retains task-critical quasi-identifiers in 6 of 12 rows. This residual risk is not merely an implementation bug: some facts, such as pregnancy, sexual orientation, domestic violence, or minority status, can be both sensitive and necessary to answer the domain question.

8 Qualitative Analysis

In `clinical_0006`, the original vignette describes rheumatoid arthritis with MCP joint swelling, positive anti-CCP, and methotrexate 15 mg weekly plus folic acid 1 mg daily. The privacy-first output replaces these with “finger joint areas,” “a specific antibody test,” “an

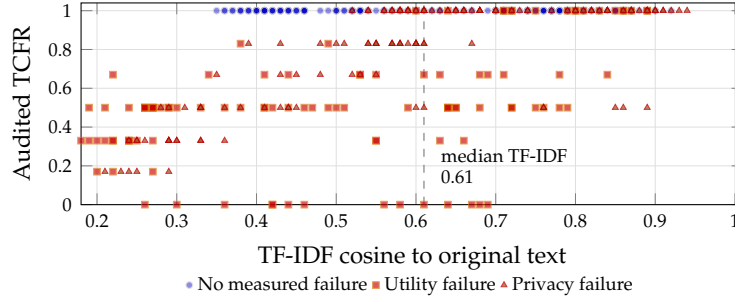


Figure 2: Surface similarity failure scatter for the 150-example non-oracle run. Among the 225 rows at or above the median TF-IDF cosine, 73 still have privacy failures and 34 have audited fact loss.

Method	Direct leak	QI risk	TCFR	QA
Regex	0.08	1.78	0.99	0.99
Presidio	0.49	2.49	0.68	0.79
Direct-span oracle	0.00	2.81	1.00	1.00
Privacy-first oracle	0.00	0.01	0.45	0.34
CSG oracle	0.00	0.23	1.00	1.00

Table 6: Local n200 diagnostic expansion. These local/oracle baselines are not the headline non-oracle result; they test whether the qualitative frontier persists under a larger deterministic diagnostic slice.

282 autoimmune joint condition,” and “a disease-modifying medication,” so both clinical QA
 283 questions fail. Critical Span Guard redacts the patient, clinic, date, MRN, phone, email,
 284 clinician, city, and exact age while preserving the lab, diagnosis, medication, and doses.

285 In legal_0006, generic prompting leaves application number 37645/97 and the exact
 286 filing date while preserving Article 6. The privacy-first output removes those identifiers but
 287 replaces Article 6 with “a specified article,” breaking the legal-basis question. Critical Span
 288 Guard removes the application number, name, country, institution, nationality, and date
 289 while preserving Article 6 and the reasonable-time civil-proceedings claim.

290 The remaining frontier is visible in legal_0005: generic and privacy-first prompting avoid
 291 the measured QI hits by generalizing away the specific section 42 Fair Employment claim,
 292 while Critical Span Guard preserves Article 6 and the exact claim. The preserved statute
 293 name still contains “Northern Ireland” and “1976,” so the row remains a measured QI-risk
 294 case. This is a useful limitation: some legal statute, jurisdiction, regime, or offense specificity
 295 may be both important for legal reasoning and quasi-identifying. A residual audit of the
 296 eleven remaining CSG QI rows found six clear task-critical legal-overlap cases and five
 297 additional frontier candidates.

298 These examples motivate a policy distinction that span-level metrics cannot make alone.
 299 Some outputs are clearly bad because they leak direct identifiers. Some are clearly bad
 300 because they remove the answer. Others are frontier cases where the correct decision
 301 depends on the intended release setting. A private internal evaluation corpus may preserve
 302 more legal specificity than a public release. A public synthetic-data release may need to
 303 generalize facts that are otherwise useful. Our metric does not resolve these policy choices,
 304 but it makes them visible.

305 9 Limitations and Ethics

306 The clinical split is synthetic. The legal split is public TAB-derived text, but legal fact and
 307 claim extraction is heuristic and manually spot-checked rather than fully expert annotated.
 308 Audited TCFR uses an LLM judge; it is useful for separating paraphrase from omission,

Method	Direct leak	QI risk	Exact TCFR	QA
Regex	0.08	2.17	1.00	1.00
Presidio	0.67	2.67	0.82	0.83
Direct-span oracle	0.00	2.50	1.00	1.00
Privacy-first oracle	0.00	0.00	0.61	0.53
CSG oracle	0.00	0.67	1.00	1.00

Table 7: Handwritten robustness stress slice. This diagnostic slice is separate from the non-oracle headline run.

but independent annotations have not yet returned and manual review remains necessary for final paper claims. The privacy metrics detect known direct spans and heuristic quasi-identifiers; they do not provide k-anonymity, differential privacy, legal compliance, or protection against a motivated adversary with external knowledge.

The intended use of this work is evaluation and safer data transformation research. We do not release or process real protected health information. The main ethical lesson is that aggressive redaction can make sensitive data safer to share only by making it useless or misleading. Responsible release pipelines should therefore state the downstream utility requirements explicitly and audit both residual privacy risk and task-critical meaning loss.

10 Conclusion

De-identification systems should be evaluated not only by what they remove, but also by whether they preserve the facts needed for downstream reasoning. On a controlled clinical/legal benchmark, generic anonymization leaks identifiers and quasi-identifiers, privacy-first anonymization often removes the answer, and a small amount of task-aware structure improves the observed privacy–utility tradeoff. The stress slice further shows that some privacy–utility conflicts are inherent: the sensitive fact may also be the task-critical fact. This makes task-critical utility evaluation a necessary complement to span-level privacy metrics for sensitive text transformation.

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